

DATABASE MANAGEMENT SYSTEMS

UNIT 1 — Assignment Answers

Section A: True / False and Multiple Choice

1. Data are known facts that can be recorded and have implicit meaning.

Answer: True

2. A record is a collection of related data items.

Answer: True

3. In a table, rows are called attributes.

Answer: False — rows are called tuples or records.

4. Columns in a table are also known as fields or domains.

Answer: True

5. A database is a collection of unrelated tables.

Answer: False — a database contains related data/tables.

6. Different users can have different views and access levels to the same database.

Answer: True

7. All data items in a single attribute must be of the same data type.

Answer: True

8. A DBMS includes both the database and programs to access the data.

Answer: True

9. Without rules and regulations, it is still easy to maintain database integrity.

Answer: False

10. Database systems are designed mainly for small amounts of information.

Answer: False

11. A DBMS must protect data against system crashes and unauthorized access.

Answer: True

12. One purpose of a DBMS is to avoid anomalous results when data are shared by several users.

Answer: True

13. A DBMS is described in the notes as a computerized record-keeping system.

Answer: True

14. Online retailers do not typically use databases.

Answer: False

15. Telecommunication companies use databases to keep records of calls and generate bills.

Answer: True

16. In file-processing systems, the same information may be duplicated in several files.

Answer: True

17. Data redundancy can lead to data inconsistency.

Answer: True

18. In file-processing systems, writing new application programs to retrieve scattered data is easy.

Answer: False

19. Integrity constraints may require that certain values, such as account balances, never fall below zero.

Answer: True

20. Concurrency problems can arise when multiple users update data at the same time.

Answer: True

21. What is the basic building block of any DBMS?

Answer: A. Data

22. Which term refers to facts, figures, or statistics with no particular meaning?

Answer: C. Data

23. What is a collection of related data items called?

Answer: B. Record

24. In a table, what are the rows called?

Answer: C. Tuples

25. In a table, what are the columns called?

Answer: B. Attributes

26. What is a collection of related records called?

Answer: C. Table

27. What is a collection of related tables called?

Answer: B. Database

28. Why are tables T1 to T4 considered a related collection?

Answer: B. They share common attributes

29. What is the primary goal of a DBMS?

Answer: B. To provide convenient and efficient data storage and retrieval

30. In a database, data is organized strictly in which format?

Answer: B. Row and column

31. In a row, the data items may belong to

Answer: B. Different data types

32. In a column (attribute), the data items are

Answer: B. Of the same data type

33. Which of the following best defines a DBMS?

Answer: B. A collection of interrelated data and programs to access those data

34. What is the role of the management system in a DBMS?

Answer: B. To provide rules to maintain the database and its integrity

35. Which of the following is NOT a typical application area of databases listed in the notes?

Answer: C. Weather forecasting

36. Which application area uses databases for reservations and schedule information?

Answer: B. Airlines

37. Which application area uses databases for customer accounts, loans, and transactions?

Answer: A. Banking

38. In a university example before DBMSs, information was stored in

Answer: B. Operating system files

39. Which of the following is a disadvantage of file-processing systems mentioned in the notes?

Answer: B. Data redundancy and inconsistency

40. Which term describes the requirement that a funds transfer must happen in its entirety or not at all?

Answer: C. Atomicity

Section B: Essay Questions

41. Define a database and explain its main purpose in an organization.

A database is an organized collection of related data that can be stored, managed, accessed, and updated efficiently. Its main purpose in an organization is to provide a reliable way of storing and retrieving information. A database helps an organization keep information organized, reduce unnecessary duplication, support data sharing, and make information available to authorized users when needed.

42. Discuss the advantages of using a database management system (DBMS).

A DBMS provides several important advantages:

1. Efficient data storage and retrieval — users can store and find information quickly.
2. Reduced data redundancy — unnecessary duplication of the same information can be reduced.
3. Improved data consistency — reducing duplication helps ensure that stored values remain consistent.
4. Data sharing — multiple authorized users can access the same database.
5. Security — access controls can restrict users according to their permissions.
6. Integrity — rules and constraints help keep data accurate and valid.
7. Concurrency control — the DBMS helps manage situations where several users access or update data at the same time.
8. Recovery — the DBMS can help protect data against system failures and crashes.

43. Explain the different views of data in a database system.

A database system can provide different views of the same data for different users. An external view is the part of the database that a particular user or group needs to see. The conceptual view describes the overall logical structure of the entire database, including its entities, relationships, and constraints. The internal view describes how the data is physically stored in the computer system. These levels allow users to work with data without needing to know all the physical storage details.

44. Describe the difference between data and information with examples.

Data are raw facts or figures that can be recorded. For example, 85, 90, and 75 could be marks recorded for a student. Information is data that has been processed or organized so that it has meaning. For example, calculating those marks to determine that the student's average is 83.3 provides useful information. In simple terms, data are raw facts, while information is meaningful or processed data.

45. Discuss the role of databases in modern business organizations.

Databases are important in modern organizations because they provide a structured way to store and manage large amounts of information. Banks use databases for customer accounts, loans, and transactions. Airlines use them for reservations and schedules. Telecommunication companies use them for call records and billing. Universities use them to manage student and academic information. Databases also support data sharing, security, reporting, efficient retrieval, and consistent record keeping. They therefore help organizations manage information and support their daily operations.

46. Explain the three-level architecture of database systems.

The three-level architecture consists of the external, conceptual, and internal levels.

1. External level — this is the user view. Different users can have different views of the same database depending on what information they need and what they are authorized to access.
2. Conceptual level — this is the overall logical view of the database. It describes the complete structure of the database, including data items and relationships, without focusing on physical storage.
3. Internal level — this is the physical storage view. It describes how the data is actually stored and organized on storage devices.

The three levels separate user views from logical database design and physical storage.

47. Describe the importance of data independence in database management systems.

Data independence means that changes at one level of a database system can be made without requiring major changes at other levels. It is important because it reduces the effect of changes on applications and users. For example, changes to physical storage should not require users to change the way they view the data. Data independence therefore makes database systems easier to maintain, modify, and manage.

48. Discuss the advantages of sharing data through a database system.

Sharing data through a database system allows multiple authorized users and applications to access common information. It reduces the need to maintain separate copies of the same data, which can reduce redundancy and inconsistency. It also improves cooperation between departments, supports faster access to current information, and allows organizations to make use of a common source of data. Access controls can also ensure that users only see or change information for which they have permission.

49. Explain how databases improve data security and reduce redundancy.

Databases improve security by allowing organizations to control who can access particular data and what actions users can perform. Different users can be given different views and access levels. Databases reduce redundancy by allowing related information to be stored and managed centrally rather than being duplicated across many separate files. Reducing duplication also helps reduce the possibility of inconsistent copies of the same information.

50. Describe the external, conceptual, and internal views of data in a DBMS.

The external view represents the specific information seen by an individual user or group of users. Different users may therefore have different views and access levels. The conceptual view represents the complete logical structure of the database, showing how the data is organized and related. The internal view represents the physical organization of the data in storage. Together, these three views separate what users see from the logical design and the physical storage of the database.